

Cover & Welcome to BUITEMS

Institution: Balochistan University of Information Technology, Engineering & Management Sciences (BUITEMS). Core Value Statement: Quality & Excellence in Education. Document: Undergraduate Prospectus 2024, Faculty of Information and Communication Technology.

Global Recognition: Global HUB for SDG8, United Nations Academic Impact (2018- 2021-2024). Institutional Rank: Ranked No. 4 by the Higher Education Commission for Computer Science and Information Technology. International Networks: BUITEMS is a member of 6 reputed international networks and

associations. Graduate Outcomes: 80% job placement ratio within the first year after graduation. Campus Life: 100+ open days and events throughout the year. Come to BUITEMS on an event day to explore the university yourself and see who we are and what we do. Website: <http://www.buitms.edu.pk>

Welcome Message from Dr. Khalid Hafeez

I have great pleasure in welcoming you to the BUITEMS. We pride ourselves on being a university for delivering quality education that is supported by excellent research faculty and infrastructure in the region. We aim to produce new knowledge through research and practice that has local, regional and global relevance and impact. At BUITEMS we mix learning with an emphasis on real-world experience and 'Future of Work'

outlook. For industry our laboratories are equipped with the latest technologies. For businesses and entities our academic faculty offer a range of high-quality research, consulting and short courses. Our class room teaching is informed by research and global practices. We support student learning with guest lectures, workshops, training courses, career seminars and employability skills. Therefore, our graduates possess global vision and are able to provide solutions for business, social and industry problems.

We offer excellent national level sport facilities, high quality accommodation and transport services. Our proximity to the Quetta Airport enhances our connections and access to national and global opportunities. Under the guidance of the Honorable Governor of Balochistan as the Chancellor, we take pride in delivering support and guidance to enable our students to become successful professionals and global citizens. I look forward to working with you.

Dr. Khalid Hafeez Professor of AI and Digital Transformation Fellow Royal Society of Arts (RSA-UK), Fellow Chartered Management Institute (CMI-UK), Senior Fellow Higher Education Academy (SFHEA - UK), Certified Six Sigma Master Black Belt (SSMBB-US), Certified European Foundation of Quality Management

(EFQM-Brussels-Belgium), Certified Lean systems and Process Manager (Herrington

Institute -US), Certified Project Manager (Herrington Institute -US), Certified Natural Linguistic Programming (NLP-UK).

BUITEMS Explained The BUITEMS Advantage: Wide Range of Programs

We are committed to provide quality education with a focus on research and to equip students with the art of living as productive members of the society, contributing to the socio-economic uplift of Pakistan in general, and Balochistan in particular. Before getting started let's explain the difference.

Our Programs

BUITEMS transformative education provides a wide range of academic programs spanning comprehensive areas of engineering, sciences and arts. It also provides multidisciplinary and cross-faculty learning opportunities.

Flexibility and Exploration

BUITEMS offers a diverse and comprehensive choice of courses in five faculties with a range of undergraduate and graduate programs. We motivate our students to explore their interests, discover their talents and pursue their passions, and to create their own BUITEMS experience!

BUITEMS Structure: Who Does What?

BUITEMS quest for academic excellence is based on five fundamental pillars; the students, the faculty, curriculum, the learning environment and the civil society we serve. We stand committed that in our academic offering we shall not only follow the standards, we shall also set new "Records of Excellence."

The University: There are five faculties and each faculty is comprised of multiple departments. The Faculty of Information and Communication Technology (FICT) has seven departments: Computer Science, Information Technology, Computer Engineering, Electronic Engineering, Electrical Engineering, Software Engineering and Telecommunication Engineering. The university through the office of the Registrar,

Controller of Examinations and the Directorates helps in maintaining a conducive environment for teaching and learning at BUITEMS. The Faculties: The Faculty and its Departments are responsible to: Determine the curriculum Organize lectures, seminars, practicals and projects

Set and mark examinations Maintain academic excellence and professionalism Provide an environment conducive to learning, teaching, academic inquiry and innovation. What are we looking for?

We are looking for individuals who can make it to be great future contributors in the fields of science, engineering and arts. We strongly encourage all applicants to choose fields of study that truly interest them. With the brilliant faculty at FICT and students with their best interest in a given field, we are confident that we will make great scientists and engineers out of our

students. We are looking for students who: have a strong academic ability and potential have a strong interest in the fields they are applying for will benefit from and grow in the BUITEMS environment We strongly advise you to think carefully when you are choosing your field of study. You should

not choose a field of study just because you think you ought to do, think about a field of study you are most excited and inspired about as you will be studying the field for the next few years. Academic Ability and Potential: We are looking for students with a compelling academic background and potential. This should be reflected in all the past qualifications and the entry test. BUITEMS expects its students to excel further in terms of academic

performance. Therefore, you should be aiming at excellent cumulative grade point average (CGPA) throughout your degree program. According to our experts, keeping up closely with all the instruction and independent study throughout the semester helps you perform brilliantly. Interest in Your Field of Study: At BUITEMS, we are looking for students who are truly

excited with the field of study they have chosen and are not just interested in the degree at the end. We firmly believe that the academic excellence can only be achieved when the student is stimulated by his or her field of study; we believe that learning comes through stimulation and interest. Self Motivation: We are looking for students who are self motivated and who can think critically. We are looking for students who can understand the model of the higher

learning where a student is required to study and work independently in addition to classroom learning. If you believe you can excel in higher learning you need to be self- motivated, self-disciplined and have a desire to learn the breadths and depths of knowledge in your field of study. If you think you have these abilities, we're looking for students like you and hope you will apply. See Applying to BUITEMS for further

information. Applying to BUITEMS: Four Simple Steps

1. Major (or field of study): You will be studying a particular field (also called major) for several years. Therefore, you should make sure that you choose a field that interests and excites you.

2. Apply: Fill out the admission form available by visiting admissions.buitms.edu.pk or the admissions office at campus. Submit the form online or at the admissions office along with the requisite documents and bank receipt of admission processing fee.

3. Calculating the Merit: Merit is based on the cumulative of 17% for the marks scored in matriculation examination, 50% for the marks scored in intermediate examination and 33% for the marks scored in the NTS admissions test. 4. Test: Eligible students are invited to the NTS admissions test. The test is conducted on

the NTS standard test for a given major. 5. Admission Offer: Admission will be offered based on the merit calculated from the marks scored in matriculation, intermediate and the NTS admission test. International

Students

We, at BUITEMS, believe that diversity brings innovation. We highly encourage international students to apply for admission at BUITEMS. Many international students from different parts of the world are enrolled in various study programs at BUITEMS. The eligibility requirements for international applicants are the same as for the applicants from within Pakistan. An applicant for

transfer from a local or foreign institution is required to have passed the BUITEMS admission test by securing equal or more marks than the minimum merit of the program in which the student seeks admission. However, acceptance of request for transfer will depend on availability of seats, and the quality of academic work already done by the applicant. As a BUITEMS student, you will be part of one of the nation's largest engineering, sciences and art schools. BUITEMS is a community known for its excellence with a student body of more than

8,000 and faculty from the world's leading institutions. You are advised to look up full details on individual field of study, credit hours and research aspects so as to get the maximum out of your chosen field of study. How do I apply?

You can apply through online BUITEMS admissions system. Download and print the application form. Attach all relevant documents and the receipt of application processing fee with the application. Send the complete application to the BUITEMS admissions office, make sure that the hard copy of your application

reaches the admission office before the closing date.

What are the entry requirements?

Entry requirements for undergraduate and graduate programs are available in the Disciplines and Departments section of this prospectus. Supporting Documents Required

1. Attested copy of Secondary School Certificate

2. Attested copy of Higher Secondary School Certificate 3. Attested copy of the Applicant's CNIC/B-form 4. Attested copy of Local/Domicile 5. Bank draft/pay order/receipt of cash payment (admission processing fee) 6. Six recent photographs (passport size)

7. Attested copy of CNIC of the applicant's father/guardian

8. Attested copy of Character Certificate from the last institute attended What are the payment options?

Online application processing fee is Rs. 4000 for national applicants and US \$55 for foreign

applicants. Whereas the fee for manual application processing is Rs. 3500. Please note that these amounts are non-refundable and must be deposited on or before the last date of the application submission. You can avail one of the following options to deposit the application processing fee: Bank draft/pay order drawn in favor of Registrar BUIITEMS, Quetta.

Cash deposited in Account Number: 23587000000201, Habib Bank Limited, BUIITEMS University branch, Quetta, Pakistan. Where to send the documents?

Post your supporting documents and hardcopy of the Admission Form along with bank draft /

pay order (admission processing fee) to the following address: Address: Admissions Office, Balochistan University of Information Technology, Engineering and Management Sciences, Takatu Campus, Airport Road, Baleli, Quetta. UAN: +92 81 111-717-111 Other lines: +92 81 2880560 / 2880136 / 2880140 / 2880386 / 2880432 / 2880527 /

2881076 / 2880511 Extensions: 163, 216, 217 What next?

Applicants will be called to take the admission test. The admit card for the test will be issued. In

case of non-receipt of admit card, the admission office may be contacted. Please note that only candidates with complete applications will be notified. Candidates without the admit card will not be allowed to take the test/interview.

Frequently Asked Questions What is the duration of a program at BUIITEMS? The duration of BS programs is four years, equally divided into eight semesters. However, the B. Arch program is spread over

five years (ten semesters). Each semester is 18 weeks long: 16 weeks for teaching, and two weeks for the conduct of midterm and final examinations. The duration for completion of the MS program is 2 years from the date of enrolment into the MS program. Can I change my program of study after admission? Change of academic program is generally not encouraged. However, it may be allowed on the recommendation of the

Chairpersons concerned and approval of the Dean(s), within the first two weeks of the first semester, subject to the fulfillment of eligibility criteria and availability of the seat. The merit of the student has to be above the merit of the last student admitted into the program to which transfer is desired.

What is a probation period? Whenever a student's CGPA falls between 1.0 and 2.0 he / she shall be put on the first probation for the next semester. If the student fails to raise the CGPA to 2.0 or above, he/ she shall be placed on the second probation for the next semester. If the student who was earlier on second probation fails to raise CGPA to 2.0 or

above, he/she shall be placed on the last probation. If the student fails to raise CGPA to 2.0 or above in the last probation, he/she shall be dropped from the university rolls. For undergraduate and MS programs, a minimum CGPA of 2.0 and 2.5 respectively is required to graduate. Can I freeze a semester? Freezing is not allowed in the first semester of a program. A

student shall be allowed to apply for freezing of at most two semesters in his / her entire program of study, after the first semester. What is the limit of courses in a single semester? A student is required to take 15-18 credit hours course work per regular semester. However the Chairperson of the department may allow a maximum of 21 credit hour course work in one semester on the

approval of the Dean. What are the different financial assistance options at BUIITEMS? The BUIITEMS Fee Concession & Scholarship Policy has the following aspects: Merit scholarship (available after the first semester), Work and study program, Fee concession for needy students, Fee concession for siblings, Fee concession for dependants of BUIITEMS employees, Fee

in installments, and USAID need-based scholarship. For more information related to financial assistance you can contact the university advancement and financial assistance office.

Teaching and Learning: A World Class Provision The BUIITEMS Advantage: Academic Excellence

ICT R&D Fund: Every year, 10 innovative research projects from FICT are funded by the Ministry of Information Technology. Supervisions: Supervisors are specialists in particular subject areas. Student-Teacher Ratio: 14:1 student-to-teacher ratio. As a research-intensive institution with a commitment to provide high-quality education,

BUIITEMS ensures an academically rigorous and stimulating experience for its students. Our Graduate Studies Office is dedicated to provide assistance and a systematic mechanism for the completion of graduate programs (MS and PhD). Graduate programs at BUIITEMS aspire to meet the competitive edge by considering the global requirements and by improving students' academic competence through course work, field-based exercises and research studies that are relevant and useful in real world.

Programs are designed by keeping in view the global trends and requirements of the Higher Education Commission, and by improving the capacity in meeting the competitive edge at national and international level. Highly-qualified faculty members, equipped labs and a

structured program at BUIITEMS provide a high-class learning experience in the undergraduate and graduate programs. We are committed to provide outstanding academic programs that offer excellent teaching and learning opportunities. We are here to foster the leaders of tomorrow in Science, Technology,

Engineering and Mathematics (STEM) and arts, enabling the students to feel and experience the standards prevailing in the best universities of the world. We aspire to raise revenues from partnerships, research grants and technology transfer while strengthening our ability to foster innovative research, establish global partnerships, and enhance our contribution to the societal and technological advancement of our community.

Programs at BUIITEMS are inspired by our commitment to equip students, employers and the wider community with the knowledge, skills and ideas to become agents of a pleasant change in the global community. In addition to receiving course instruction and supervision from internationally recognized faculty, you benefit from a supportive and enriching learning experience at BUIITEMS.

How will I be taught?

You will benefit from a contemporary teaching and learning environment at BUIITEMS. Our facilities include modern lecture rooms, state-of-the-art labs, research clusters, and senior design labs as well as online access to many international databases including IEEE, Springer,

ACM, and ScienceDirect. In line with global academic trends, we equally emphasize independent and self study. You are responsible for your own studies and are expected to study beyond the class lectures and study material. Lectures: Lecture is the core teaching and learning point and also your starting point for research which lasts for a minimum of one hour and a maximum of three hours per week

for a given course. Lectures delivered by experienced faculty can be highly-stimulating. Depending on the course material, the instructor may use a whiteboard, digital multimedia (or both) complemented with oral instruction. Experiments: These are the laboratory classes. Subjects with real time applications are instructed for three hours hands-on practice on the equipment and tools; and are separately assessed.

Supervisions: Supervisions are special hours of in-depth discussions between individual students or groups with a subject or research area specialist teacher. Study Tours and Field Trips: The departments conduct study tours and field trips to industries and labs in different parts of the country. Students are facilitated to participate in different intra-university competitions and are also assisted in visiting other universities

for experience. Study tours are usually an extended form of field trips that involve outbound trips for more than a night. During the course of these tours, students are encouraged to build interpersonal skills with their batch mates through shared learning. Independent Research (Final Year Projects and Theses): Undergraduate students are engaged to carry out a one-year long research and development project under the

supervision of a specialist faculty member. The students usually conduct this work in the

Senior Design labs. The graduate students conduct independent research work in the form of a thesis under the guidance of a thesis supervisor. Peer Mentoring: Student volunteer tutoring is available in the form of peer-mentoring. Select senior students who are young gurus in a given field of study are enrolled as

mentors in the peer-mentoring program. The peer mentors guide and counsel students who require extra tutoring. At FICT, we provide support, encouragement, and advice to students who are facing problems related to academics or life at campus. Internships: Internships offer the opportunity to apply the

knowledge gained from academic studies in a practical workplace setting. These experiences expose students to

various occupations and industries. With a clear purpose to complete specific projects, internships give students the chance to evaluate and reflect upon their chosen career field, providing a significant source of self-awareness and skill identification. Over 400 students at FICT are directly placed under the internship program every year; while many students get advice and indirect assistance in finding internship placements.

Facilities and Resources: Support Along the Way

We are committed to connecting you with the right resources, from dedicated teaching staff to a world-class library and a range of co-curricular facilities. At BUIITEMS, we spend millions on our equipment and academic support services to create a quality environment where you feel stimulated and strengthened. Libraries: With over 40,000 books the BUIITEMS central library continues to add to

student's passion for reading. Both the campuses house one central library each with a number of small departmental libraries. The libraries also provide our students with free access to a world of books through digital library access. The library experience is augmented through a state-of-the-art automated library management system that assists in searching and locating books in the library anywhere from the campus.

Digital Library Access: Institute of Electrical and Electronics Engineers (IEEE): The IEEE/IET Electronic Library (IEL) provides access to almost a third of the world's current electrical engineering and computer science literature, featuring high-quality content from the Institute of Electrical and Electronics Engineers (IEEE) and the Institution of Engineering and

Technology (IET). Springer Link: Springer Link provides access to 503 full-text Springer Verlag Journals and 738 full-text journals formerly published by Kluwer Academic Publishing, one of the world's leading information services for Science, Technical and Medical journals. Association of Computing Machinery (ACM): The ACM Digital Library provides online access to thirty magazines and journals in computing and IT, with a complete archive

reaching back to the 1950s. It also includes the ACM Special Interest Group newsletters and conference proceedings, many with full archives. The ACM Online Guide allows ACM users to expand their searches to include non-ACM works in their results.

Research Center and Senior Design Labs: BUIITEMS is committed to provide a conducive environment to the entire learning experience of students. The research center located in the Sir Syed block is a cutting-edge facility to serve that commitment. The research center has dedicated research desks for graduate students. The research desks

are equipped with high performance computing and related resources necessary to carry out graduate research. To cater to the needs of undergraduate students, the facility provides senior design labs. These labs offer dedicated spaces for final year projects, equipped with computing and other necessary resources tailored to the students' disciplines within ICT.

Art Galleries: Thanks to the Faculty of Basic Sciences, the campus boasts several art galleries. These galleries showcase a fascinating collection of modern art, including artifacts, paintings, and sculptures created by the students and faculty of FBS. Even if not directly related to your course, students of FICT

spend time and receive their share of learning and knowing arts at the galleries.

Online Learning: Uninterrupted Education During the

Pandemic The pandemic did not halt our progress; neither did we. In the wake of COVID-19, BUIEMS

decided to continue the academic activities online and remained one of the few universities in Pakistan with a 100% readiness to continue the classes online. BUIEMS Response to Pandemic COVID-19

COVID-19 has halted global, social, and economic progress challenging 21st Century lifestyles

and workplace practices, thus impacting conventional institutional arrangements. As an immediate response to the pandemic, understanding the gravity of the situation and its consequences, BUIEMS leadership started with the closure of all its facilities for faculty, students, and staff, while ensuring that the learning process and research activities continued. BUIEMS stands together with the global community during COVID-19 and plays its role in

continuing learning processes and research; supporting government with policy frameworks; encouraging other educational institutions to continue online education and supporting them in the process; and collaborating with national, regional, and international educational institutions. BUIEMS Agreement with Microsoft

In order to create a virtual work space, BUIEMS signed an agreement with Microsoft and

issued accounts to all students and employees. Microsoft Office 365 (O365) accounts were acquired, and training on the effective use of Microsoft Teams (MT) for online classes was imparted to all Deans and Department Chairs. BUIEMS realized that internet connectivity, cost, and adaptability to the new working environment would be challenging and would require a resolve for the successful conduct of

online classes, especially when such arrangements had never been practiced before. BUIEMS realizes that COVID-19 will disappear; however, online teaching will not. The pandemic has

changed the way students study and the way teachers teach. Therefore, BUIEMS' focus is on the adaptability of the best new working practices, with an emphasis on creating online classes that accommodate learners. Student Testimonial:

"What impresses me more about BUIEMS is the high standards that it holds for quality, objectivity and the growth opportunities that it provides to talented and hardworking people." — Babar Ali, Graduate Student, BS Electronic Engineering

Student Life: More Than Just Studying Life at BUIEMS is a work-hard-and-have-fun culture. As members of the BUIEMS family, the

students, faculty, and staff enjoy an exciting, vibrant, and colorful life at BUIEMS. From high- class academic lectures and laboratory experiments to exciting sports, art events, and service projects, there are always great things happening on campus. The BUIEMS Advantage: Work Hard and Have Fun

Student Organizations: 20+ student clubs and societies to choose from. **World-Class Sports Facilities:** BUIEMS sports complex and stadiums are among the best in the country. **Arts and Culture:** Non-stop events complemented with excellent venues. **Alumni Testimonials:** "Quality education and learning...great

campus and above all lively environment." — Isma

Batool, Alumna, BS Software Engineering "I found it to be the best institute...full of Opportunities for some of the most hardworking students around!" — Anil Amjad, Alumna, BS Computer Science Events and Fests

From the one-of-a-kind cultural festivals on campus to accomplished student performances in theater, arts, and culture, Student Affairs plays a leading role in life at BUIITEMS. Student Affairs comprises more than twenty student clubs and organizations that continuously provide fuel for a constructive social life at BUIITEMS. Student Affairs, on average, conducts at least one event each day on campus. The scale of the events varies from an attendance of 50 participants to

thousands of participants and spectators in flagship events like The Spring Festival, Culture Fest, and the Homecoming. The preparations for the events and the event days create an amazing atmosphere on campus where every student and staff member seems to be working with an exceptional bond. The sense of ownership of the institution among the students and staff is exemplary at BUIITEMS. The quarterly newsletter and Markhor magazine provide coverage of the events and

developments on campus.

The Olympiad

The annual Olympiad marks the culmination of athletics and sports at BUIITEMS. The Olympiad is a week of non-stop sports and athletic competitions in more than fifteen sports categories, with over one thousand athletes participating from among the staff and students. The Olympiad

is witnessed by thousands of spectators and sports fans from within campus and the city. Community Service

Another important aspect of life for the BUIITEMS family has been community involvement and service. BUIITEMS family members have a very special motivation for social service and social

contribution to society. The students and staff are contributing to the elementary education of child laborers under the Free Citizen Schools. This organization was formed by the staff and students of BUIITEMS. Through the organization, the students and staff of BUIITEMS have been directly educating young children who are working to meet the finances of their families. The children are educated in basic reading, writing in Urdu and English, and basic math and science.

The students and staff also visit special community schools on weekends in different parts of the city and educate students using modern instruction methods. The students and staff are also helping poor and needy citizens who are in need of support with their health problems. BUIITEMS blood donation drives have earned good recognition in the province, and a large number of citizens have been helped with BUIITEMS blood donations. BUIITEMS has also been

helping the citizens of the province in their career planning, and providing counseling on scholarships. The counseling and trainings are organized for the young graduates of different universities of the province. Continuous Buzz

BUIITEMS enhances the ability of students and staff to reach their fullest potential through diverse academic, personal, and professional-development experiences. To achieve this, we foster a welcoming,

stimulating campus life environment where students and staff develop intellectually, experience meaningful co-curricular opportunities, evidence civic responsibility, model intercultural and interpersonal understanding, and promote health and well-being. The

BUITEMS family is prepared to freely pursue life-long personal and professional fulfillment, engagement, and stewardship for ever-changing local and global communities. Sports: Get Ahead of the Game

Whether you're a world-class athlete or new to exercise, we have the facilities and expertise to keep you motivated. From the fun to the competitive side of sports, we love it all at BUITEMS.

We have more than 3,000 members utilizing our sports facilities, and approximately 1,000 students participate in different sports tournaments inside and outside the province. With so many activities to try out and plenty of post-exertion socializing opportunities available, you can get fit and have fun at the same time. If you are a talented athlete in training, we offer a range of

services to support you as well as sports bursaries and funds for team participation in inter- university championships and tournaments. Sports Facilities Breakdown

BUITEMS boasts providing one of the best sports facilities in the nation. Sports and athletics form one of the core features of life at BUITEMS. The newly constructed sports complex is the jewel of recreation facilities at BUITEMS. The sports complex provides venues and gear for a range of sports including basketball, handball, badminton, table tennis, and volleyball. The fully equipped gym and fitness center are part of the Sports Complex that houses the finest fitness

equipment to keep you agile and healthy. The world-class cricket stadium at BUITEMS is a commendable facility for cricket lovers. The stadium is operated with the support of the Pakistan Cricket Board (PCB) and regular regional tournaments are held at the stadium. The football ground at BUITEMS is no less of a treat for football fans and players. The lush green ground remains one of the busiest sporting facilities

on campus around the year. In addition, a number of separate facilities for basketball and other sports are located in different parts of the campus. Get registered and enjoy the excellent facilities to cope with your tedious work and study routine. The University Features: Multi-purpose sports complex, high-class gym, PCB standard cricket stadium, high-class football ground, and basketball courts.

In and Around BUITEMS

BUITEMS is located in Quetta, also called the fruit basket of Pakistan. Quetta is one of the most beautiful cities of Pakistan distinguished by a unique backdrop of mountains, beautiful lakes, and fruit orchards in the outskirts. The city is famous for its amazing weather and beautiful fruit orchards in and around. Quetta is surrounded by small valleys and hills. The juniper forest in Ziarat, located at about 125 kilometers north of Quetta, is another natural distinction, being the

second-largest juniper forest in the world. Located in Pakistan's most attractive destination, BUITEMS has loads to offer you as a student. The World in BUITEMS

With students and faculty from different cities around Pakistan and the world, BUITEMS brings

a diverse world to the campus. With such a unique blend of people, life in the university is never dull. Apart from the fervent exchange of academic ideas, a host of programs and activities are offered. Events like the spring Olympiad and cultural show allow students to showcase their country's culture and share the festive

mood and vibes with the BUITEMS community.

Campus Tours

A frequent sight at BUITEMS is that of young school children being escorted by a guide, walking across different facilities on campus. Young school children from different schools of the province are invited to the campus for a university life orientation. They are given an orientation of the variety of science, engineering, and arts disciplines offered at the university. The children

spend a day at the campus and witness university life. This unique experience at BUITEMS for school children is aimed at helping students plan their careers and futures early in their lives. This exercise will help produce professionals in the province and in the country who will truly excel in their fields of specialization.

Restaurants and Cafeterias

Home is where the heart is: from food carts to fine dining, the University is full of places with good eats, drinks, and treats. The tea and food houses of BUITEMS serve the best. Retail dining options include Road Stoves, Foodie, Usmania Café, Mid Canteen, Café Chromepur,

and Rooftop. BUITEMS dining recognizes the great power and importance of food. Dining rooms are gathering places, and breaking bread together helps create a sense of community and comfort. You can easily drop into the cafes and place an order of your choice. You can also join the free-wheeling academic debates, poetry recitations, and literary repartees in the cafes.

Entertainment

At BUITEMS, we deliver fun everyday, at every milestone, every challenge, and every achievement around the year. There are many collaborative competitions and celebrations organized to make you feel like part of the family. Each day on campus there is an event engaging the BUITEMS family and visitors from outside. The scale of the events varies from the attendance of a small group of participants to thousands of participants and spectators in flash

events like the Spring Festival, Culture Fest, Olympiad, and Homecoming. The preparations for the events and the event days create an amazing atmosphere on campus where every student and staff member seems to be working with an exceptional bond. Airport and Highway

BUITEMS is located just three kilometers away from Quetta International Airport. The airport,

through many national and international carriers, provides a round-the-clock connection to other major cities of Pakistan and abroad. The campus is located adjacent to the RCD International Highway, which provides road access to the rest of the nation and Central Asian countries. Going Places with BUITEMS

In an interconnected world, graduates need to stand out as culturally sensitive, well-informed, and articulate individuals who can learn and think for themselves. They will also need to keep abreast of global developments in order to meet its challenges. Here at BUITEMS we encourage, support, and assist our students to participate in many national and international conferences, competitions, and study tours. BUITEMS effectively helps in fostering national and international relationships across cultures and borders through its programs.

Attractions In and Outside the City

Hanna Lake: Nestled in the hills ten kilometers east of the city, Hanna Lake is a turquoise body of water that contrasts markedly with the bare surrounding hills. It is an attractive

destination for vacationers, with facilities for boat hire. A lakeside restaurant is crowded with hikers and campers during holiday periods. **Hazarganji Chiltan National Park:** Located 20 km south-west of Quetta, this is a protected park area. In the folds of the mountains, according to legend, there are over a

thousand buried treasures—reminders of armies that once passed through the region including the Bactrians, the Scythians, the Muslims, and the Mongols. The Chiltan Hill Viewpoint in the park provides a panoramic view of the city. **Pir Ghaib:** A waterfall and picnic spot located 70 km from Quetta City in the historic Bolan Valley.

Kharkhasa: Located 10 km west of Quetta in a 16 km long narrow valley that contains a variety of flora and fauna species. **Museums & Culture:** A visit to the nearby cities of Kirani and Ziarat are popular scenic places for tourists traveling to and from Quetta. The Quetta Geological Museum, located on Sariab Road, has a collection of rocks and fossils. The Command and Staff College

Museum is dedicated to British military history, housed in the former bungalow of Field Marshal Bernard Montgomery. The Quetta Archaeological Museum, located on Wafa Road, has a collection of rare antique guns, swords, manuscripts, and a display of Stone Age tools, prehistoric pottery, and articles found in Mehrgarh. There are also coins, manuscripts, and photos of Quetta before the 1935 earthquake. The Balochistan Arts Council Library houses arts and crafts from the province.

Outing Spots: If you want to explore further afield and are looking for somewhere to relax and escape the city buzz, Hanna Lake, Urak Valley, and holiday stations such as Ziarat and Pishin are home to fresh fruits and natural forests, serving as famous outing spots around Quetta.

Special Events & Engagements **SDGs & FICT:** BUIITEMS as United Nations Academic Impact Hub for

Goal 8

The Balochistan University of Information Technology, Engineering and Management Sciences was chosen as the SDG Hub for Goal 8 (Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all) by the United Nations Academic Impact (UNAI) in 2018 for a period of three years. In recognition of the research, innovation, and scholarship undertaken in support of the SDG, UNAI extended the

term for BUIITEMS as the global SDG8 hub for 2021-2024. The Faculty of Information and Communication Technology, BUIITEMS is an active partner of UNAI, and ensures periodic retrospective implementation of the SDG agenda by incorporating and evaluating its objectives in light of the sustainable development goals. All departments of FICT ensure that student capstone projects contribute to at least one of the seventeen goals.

Further, to keep stakeholders engaged in support of the SDGs, a number of national and international events are arranged including: The Hult Prize Foundation in partnership with the United Nations; Promoting SDG8 to Advance Agri-Entrepreneurship for Balochistan Youth, in

partnership with FAO; 1st International Symposium on Building Economic Resilience against Pandemics, in collaboration with the UNDP; World Youth Skills Day in collaboration with ILO; and a dedicated track on Sustainable Development & Society during our famous International Conference on Computing, Electronics, and Electrical Engineering (ICE Cube).

Quetta Literary Festival (QLF)

The Quetta Literary Festival is aimed at promoting art, literature, aesthetics, and discussion in the referential context of Quetta and Balochistan. The festival not only seeks to create a shared space for authors, artists, journalists, thinkers, filmmakers, and musicians, as well as for

readers, audiences, and art lovers, but also brings together policymakers at the national and provincial levels to discuss challenges and seek solutions in an interactive & celebratory environment. Hence, QLF is an avenue where thought and deliberation meet zeal and excitement. QLF significantly impacts the student experience by focusing on various dimensions, including youth

engagement, cultural and intellectual representation, promotion of arts and aesthetics, and long-term constructive development. It also fosters investment in the province of Balochistan. For details please visit www.quettalf.org. International Conference on Computing, Electronics, and Electrical

Engineering (ICE Cube)

The ICE Cube conference is a renowned international conference hosted by FICT biennially. The conference focuses on the exchange of ideas on burning problems of national importance such as e-governance, energy, information security, communication security, mining of public- sector data for meaningful analysis and policymaking, e-education, and sustainable

technologies. The ICE Cube was first held in 2016, followed by 2018 and then 2021. The 3rd International Conference on Computing, Electronic and Electrical Engineering (ICE Cube 2021) was held on 26th and 27th Oct 2021 in Hybrid Mode (i.e., physical and virtual). Further details of the IEEE ICE Cube are available on the conference website: <http://www.icecubeconference.org>.

Student Achievements

Looking back, these years have been etched into the history of BUIITEMS not as periods of tragedy but as times of resilience and growth. With a progressive mindset, we navigated through difficult times.

Alhamdulillah, the Markhors achieved many noteworthy successes during those years. Participation in the Bali Asia International Model United Nations (BAIMUN) in 2020:

Our students are our pride. Atqa Rana, a student of Software Engineering, was the only delegate selected from Pakistan to participate in BAIMUN, held in Indonesia. She was selected as a delegate of Pakistan in the UNWTO (United Nations World Tourism Organization) council under the topic of Sustainable Tourism, along with a position paper, and proposed a UNWTO resolution with the agreement of other countries.

BUIITEMS Students Shine Bright in Huawei Seeds for the Future Program-2020: Technology is the new gold, and nations that are investing in becoming technologically sound are securing their future. A team of students from the BS Computer Engineering program secured top positions in the Huawei Seeds for the Future Program 2020.

Team BUIITEMS Secured a Position Among the Top 10 Global Winners of the Google Android Developers Challenge: Blessed are those who help others with solutions that are out of the box. Team BUIITEMS from the Faculty of Information and Communication Technology was among the top 10 winners of the ADC. The team created an app named "AgriFarm" that helps farmers detect plant diseases and prevent major

damage in fruits and vegetables such as tomatoes, corn, and potatoes. Research & Funded Projects

Research and development forms the core of life at Balochistan University of Information Technology, Engineering and Management Sciences (BUIITEMS). As an institution of higher learning, BUIITEMS is increasingly focusing its efforts and energy on R&D.

The BUIITEMS Advantage: Innovation and Commercialization

To this end, research groups at BUIITEMS are continuously busy solving problems pertinent to the province, the nation, and the world. Research at BUIITEMS is rapidly evolving to a level where it will inspire game-changing ideas and new technologies that will help drive economic growth, while improving human life on both a regional and global scale.

This is all possible because the culture at BUIITEMS encourages thinking big and the fearless pursuit of grand challenges. Another factor that sets BUIITEMS apart from other research universities is the harmonious, highly collaborative environment among the faculty that fuels the university. The challenging research goals at BUIITEMS are continuously accomplished through funding from BUIITEMS and with the support of national and international agencies who share a

passion for great ideas, courageous thinking, and a desire to shape the future to come. With the courage and passion of the people at BUIITEMS and with the help of friends, BUIITEMS will achieve its greatest aspirations to become the technological research university of the twenty first century.

Office of Research, Innovation, and Commercialization (ORIC)

To manage enthusiastic and comprehensive research objectives, BUIITEMS has established a central Office of Research, Innovation, and Commercialization (ORIC). ORIC has become a pivotal entity to promote research in the university and is assisting researchers within the university to think up unthinkable research solutions and seek national and international funding for their endeavors. Through ORIC, BUIITEMS is also encouraging its researchers by providing

them incentives to publish quality research work in the most reputable research journals worldwide.

Highlighted Funded Projects

1. Project Title: The Outbreak of Online Education in the Growing Digital Economy and Digital Gender Divide Issues Total Funding: PKR 1.5 Million

Funding Source: Massey University, Auckland New Zealand 2. Project Title: Metering the aquifer using smart monitoring & data-driven approach to assist in devising adaptive groundwater management strategy in Balochistan Total Funding: PKR 53 Million Funding Source: Higher Education Commission 2020-2023

3. Project Title: Design and Prototyping of Bio-Telemetry Antenna System for the Healthcare System of Balochistan. Total Funding: PKR 2.6 Million Funding Source: ORIC Large Scale Grant (completed)

Control, Automotive & Robotics Lab (CARL)

The Control, Automotive & Robotics Lab (CARL) research team at BUIITEMS is a research cluster that has been working on the development of robot-assisted systems to solve pressing problems of Pakistan, and particularly Balochistan, using Robotics and Artificial Intelligence. The lab is part of a consortium of 11 labs in 13 universities under the name National Center of Robotics & Automation (NCRA). The lab consists of qualified Ph.Ds, industry researchers, and

experts to achieve solutions to local problems. CARL worked on a project worth 80 million PKR funded by the NCRA to research and develop a search and rescue snake robot and a pipeline of social mobile autonomous robots including various variants such as WaiterRobot, SMART, and +Rover. Established Under the National Centre for Robotics & Automation

(NCRA)

NCRA is a PSDP project worth PKR 1.295 billion under the umbrella of the HEC. The National Center of Robotics and Automation is a consortium of universities in Pakistan, headquartered at NUST College of EME in Rawalpindi. The Centre proposes to become the leading hub to think, design, innovate, and commercialize to transfer knowledge to the local economy in the area of

Robotics and Automation. It facilitates researchers in the field, helping them establish and grow the Automation industry towards Industry 4.0 and seek solutions to indigenous problems through Robotics. Key Technological Domains Worked on at CARL

Search and Rescue Robot: The hyper-redundant snake-robot, which uses state-of-the-art reinforcement learning algorithms, can maneuver in complex environments, such as collapsed buildings and coal mines, to identify human survivors. It can also be used for espionage. These robots are preferred because they can move in very narrow places, on different terrains on the ground, and in water where wheeled and legged robots fail. Our

goal is to make this robot close to a natural snake as much as possible to exploit its flexible movements. Our long-term goal is to turn our snake robot project from a prototype into an actual machine that can be deployed in earthquake-affected places and save lives by bringing state-of-the-art rescue equipment to disaster relief teams in Pakistan.

Social Mobile Autonomous Robots: Social robots can interact with humans to provide information and guidance. They have multiple applications including being stationed at airports, universities, and public gatherings. In our lab's dynamic social autonomous mobile robots pipeline, we proudly introduce the WaiterRobot, SMART (Social Autonomous Mobile RoboT), and +Rover (pronounced adrover), each contributing unique

capabilities to redefine human-robot interactions. The WaiterRobot excels in providing efficient and courteous service through advanced navigation algorithms. SMART, designed in the form of a charismatic minion, operates in theme parks and malls, engaging customers with unparalleled social finesse. Additionally, SMART and +Rover share the ability to communicate with customers, with +Rover offering an innovative

revenue-generation dimension by displaying ads. This integration of social awareness, mobility, and commercial functionality underscores our commitment to advancing the frontiers of autonomous robotics

for both entertainment and economic impact. Industrial Manipulator: Features vision-based pick & drop operations with a dynamic end-effector. It can perform its objectives even as the details of the environment change. Campus Bus: Fully autonomous driving with vision-based obstacle avoidance. It can be

deployed in unknown environments. Whegged Robot: Designed for mining exploration, space missions, and military operations, providing enhanced stability on rough terrain. Mechanical Ventilator: Designed for the COVID-19 response in times of emergency when existing facilities were not enough to fulfill the needs of the rapidly deteriorating

situation. Image Stabilization: A platform to improve the visual output of the video-feed from any camera, ranging from a smartphone app to cameras fixed on mobile robots. Lab Infrastructure and Hardware Worth

CPU Processing Server (Rs. 2.5 Million): NVIDIA DGX Server. This is a 96 times faster Deep Learning machine providing 170 teraflops of processing power. 3D Printer (Rs. 1.5 Million): Ultimaker S3. Used for rapid prototyping with high precision and custom PCB design. Motion Capture System (Rs. 2.5 Million): Utilized for cutting-edge research and the

real-time testing of complex robotic algorithms. Major Lab Achievements

Microsoft Imagine Cup Regional Winner (South Region) — Muhammad Rafi & Kabisha Anwar: This project focused on a semi-autonomous robot, WhегRoach,

capable of making smart decisions and giving an internal view to rescue teams so that

they can go safely and perform their respective tasks. The design of the robot enables it to overcome different obstacles and move stably on rough terrains. It can be used for space exploration, crop fields, SAR tasks, and mining areas to perform perilous tasks.

CARL Core Team

Dr. Anayat Ullah (Director / PI) Hamza Anwar (Research Associate) Ehtisham Ahmed (Research Associate)

Spatial Decision Support System Lab (SDSS)

The Spatial Decision Support System (SDSS) Lab is a part of the National Center of GIS and Space Applications (NCGSA). The National Center of GIS and Space Applications materializes all domains of space science, technology, and its applications, starting from space awareness & education to astronomy and astrophysics, satellite development to satellite navigation, and remote sensing & GIS applications. The NCGSA envisages working in three domains, namely,

Space Science, Space Technology, and Space Applications. Established Under the National Centre for GIS & Space Applications (NCGSA)

A substantial rise in public demand for efficient services and systems can only be met with

efficient decision support systems from the government. One way to handle this restless onslaught of demographic pressure is to improve the performance of several systems and services available to the public by making use of spatial data. The availability of data along with an effective decision support system can help in the understanding of problems and the possible outcomes of decisions.

In addition, the availability of spatial data-based decision support systems can greatly improve policy and plan-making processes while reducing time and human resources. Geographic Information Systems (GIS) is a well-known technology used to manage spatial databases. Our research group envisions contributing to the area of Spatial Decision Support Systems (SDSS) that deal with spatial dimensions through digitized geo-referenced spatial databases using geo-

spatial analysis and machine learning tools. Core Domains Covered by the SDSS Lab

1. Geo Forensic & Geo Health (including Crime Management and Health Resource Management)
2. Mineral Inventory (including Mineral Mapping and Exploration)

Major Achievements & Software Solutions

Eco-Health (Web Application): Developed for visualizing environmental variables across multiple diseases and for monitoring diseases geographically and environmentally to plan for future mitigations.

Forest Watch (Mobile Application): An application designed to monitor deforestation and tree diseases in a specific area. The Forest Department can use Forest Watch, a mobile application created by the SDSS Lab in partnership with SBBU-Dir.

Key Performance Indicators (KPIs)

Academic Research & Development: Literature surveys, technical reports, review papers, conference/journal papers, and BS/MS research thesis completion. Software Solution Development: Development of applications & software, conceptual designs, prototypes, codes, algorithm development, modeling, and simulation

tools/techniques development. Hardware Solution Development: Training kit development. Model & Maps Development: Models generation/upgradation, GIS/maps development & upgradation. Capacity Building: Space education & awareness sessions, trainings, workshops,

seminars, and webinars. SDSS Core Team

Principal Investigator: Prof. Dr. Bakhtiar Kasi Co-Principal Investigator: Dr. Shafiullah Business Development Manager (BDM): Arsalan Ali Cheema

Research Assistant: Muzaffar Ali Computer Operator: Zohaib Durrani Research Associate: Reza Tahir Research Assistant: Muhammad Amjad Computer Operator: Muhammad Talha

Academic Collaborators/Partners

1. Dr. Shabeh-ul-Hassan (University of Hamburg)
 2. Dr. Hammad Gilani (International Water Management Institute)
 3. Dr. Sajid Ghaffar (Institute of Space Technology, Islamabad)
 4. Dr. Abu-bakar (LUMS-NCRA)
 5. Dr. Laima Khan (University of Leeds, United Kingdom)
 6. National Institute of Disaster Management
 7. National Institute of Management Sciences (NIMS)
- Sectoral Collaborators/Partners

1. Provincial Health Directorate, Govt. of Balochistan

2. World Health Organization (WHO), Balochistan 3. Emergency Operations Center (EOC), Balochistan 4. Expanded Programme on Immunization (EPI), Balochistan 5. Governance & Policy Project, Balochistan
6. Forest and Wildlife Dept., Govt. of KPK 7. FIA Cyber Crime Wing, Balochistan 8. Data Command & Communication Center (D3C), IG Office, Quetta

Fields of Study (Faculty of Information and

Communication Technology) Welcome from the Dean

Information and Communication Technology is continuously restructuring our world! Welcome to the Faculty of Information and Communication Technology (ICT) at BUIITEMS, where we are committed to providing our students with an exceptional education and preparing them for success in the fast-paced, ever-changing world of technology. Our faculty is made up of experts in their fields, with years of industry experience and a passion for teaching and research. At

BUIITEMS, we offer a wide range of degree programs in some of the most sought-after fields in the world today. Our BS, MS, and PhD programs in Computer Science, Electrical Engineering, Computer Engineering, Software Engineering, Electronic Engineering, and Information Technology are designed to equip our graduates with the knowledge and skills they need to thrive in their chosen careers. Our state-of-the-art labs and facilities are equipped with the latest technology, providing our

students with hands-on experience and practical skills that are highly valued by employers. We believe that practical experience is essential to a well-rounded education, and we are proud to offer our students the opportunity to work on real-world projects that help to solve important problems and improve people's lives. Our faculty members are dedicated to providing our students with a rigorous and challenging education, but also to supporting them in their

academic and personal growth. We understand that our students are the future of the technology industry, and we take our responsibility to prepare them for success very seriously. In addition to our excellent degree programs, our research programs are at the forefront of technological innovation. Our faculty members are engaged in cutting-edge research that is solving real-world problems and shaping the future of technology. We believe that research is

an essential component of a great university, and we are proud to be contributing to the advancement of knowledge in our fields. The fields of Information and Communication Technology are vital to the future of our society and our economy. The job prospects for our graduates are excellent, with a wide range of career paths available in industries such as

software development, telecommunications, electronics, and more. Our graduates are highly sought after by employers both in Pakistan and around the world, and we are proud to have a strong record of success in placing our students in rewarding and fulfilling careers. At BUIITEMS, we believe that our students are capable of achieving great things, and we are

committed to helping them reach their full potential. We invite you to join us and become part of our community of innovative thinkers, problem-solvers, and leaders in the world of technology. — Dr. Bakhtiar Khan Kasi, Dean FICT 1. Computer Engineering

Computer engineering is a discipline that embodies the science and technology of design, construction, implementation, and maintenance of software and hardware components of modern computing systems and computer-controlled equipment. Computer Engineering at BUITEMS

At the Department of Computer Engineering, students are encouraged to hone the right mix of expertise covering both software programming and hardware design and implementation to be

able to design and utilize computing systems for industrial automation (to handle different processes and machinery in an industry to replace a human being), embedded systems (found in devices from a digital wristwatch to the space shuttle), robotics (to use machines to perform tasks done traditionally by human beings), multimedia (to combine and use multiple media formats i.e. text, audio, still images, animation, video and interactivity together), and artificial

intelligence (to design and program machines to think like humans and mimic their actions). Programs Offered & Fact File

Programs: Four-year BSCE, Two-year MSCE. Code: BSCE Duration: Four years (8 Semesters)

Accreditation: Fully Recognized by the Pakistan Engineering Council (PEC). Program Vision: To produce professional Computer Engineers who can use knowledge, technical expertise, and entrepreneurial abilities to compete with upcoming challenges in computing fields while adhering to highest standards of morality and ethics. Program Mission: Our mission is to prepare graduates to accept and fulfill

responsibilities across a broad spectrum of activities, including analysis, synthesis and design of modern computing systems, serve government and industry in various roles and contribute in academia and research. Scheme of Study (BS Computer Engineering)

Semester 1: Computer Fundamentals, Applied Physics, Calculus & Analytical Geometry, Islamic Studies and Ethics, Reading and Writing Skills (Eng.-1), Workshop Practice,

Occupational Health and Safety.

Semester 2: Computer Programming, Linear Algebra, Electronic Circuits and Devices, Communication and Presentation Skills, Linear Circuit Analysis, Pakistan Studies and Global Perspective. Semester 3: Computer Comm. and Networks, Object-Oriented Programming, Complex

Variables and Transforms, Digital Logic Design, Discrete Structure. Semester 4: Data Structures and Algorithms, Signals & Systems, Computer Architecture and Organization, Differential Equations, Management Sciences Elective - I. Semester 5: Digital Signal Processing, Microprocessors and Interfacing, Operating Systems, Social Science Elective - I, Technical Writing.

Semester 6: Database Management Systems, Digital System Design, Computer Engineering Depth Electives-I, Computer Engineering Depth Electives II, Probability Methods in Engineering. Semester 7: Multi-Disciplinary Engineering Electives-I, Software Engineering, Computer Engineering Depth Electives - III, Numerical Methods, Senior Design Project-I.

Semester 8: Social Science Elective-II, Computer Engineering Depth Electives - IV, Management Sciences - Elective II, Senior Design Project-II, Multi-Disciplinary Engineering Electives-II. Pedagogy Note: Teaching in this program is delivered through a combination of lectures and practical classes. Students typically complete three assignments and two quizzes per subject,

and some subjects also require presentations. Practical work is assessed separately. BS Program Summary

Total Credit Hours: 134 Essentials/Eligibility: F.Sc (Pre-Engineering) / ICS with Mathematics & Physics from any recognized board or equivalent with at least 60% marks, OR a Diploma of Associate Engineer (DAE) in the relevant field securing at least 60% marks.

Admission Test: Admission is contingent upon passing the admission test. Contact: Chairperson: Dr. Sibghatullah Bazai (Ext. 712) Graduate Program Information (MS Computer Engineering)

The Department offers the Master of Science (MS) degree in Computer Engineering. The Department presently has active research groups in the following areas: Computer Forensics &

GeoForensic, Communication System Group (CSG), Computational Intelligence Group (CIG), Wireless & Sensor Networks (WSN), Distributed & Networked Systems (DNS), and Information Security (IS). MS Degree Requirements: At least 9 credit hours of core Graduate level courses. At least 15 credit hours of advanced elective graduate level courses. At least 6 credit hours

for Master's Thesis. Total Credit Hours: 30. Core Graduate Courses: Advanced Computer System Architecture, Advanced Operating Systems, Advanced Computer Network. Elective Courses Domains: Computer Network and Distributed Computing, Mobile and

Semester

Wireless Networks, Parallel and Distributed Computing, Embedded Systems Security, Fuzzy Logic & Neural Networks, Artificial Intelligence and Deep Learning, Advanced Artificial Intelligence, Internet of Things, Machine Learning, Cyber Security, Cluster and Cloud Computing, Wireless Networks Security, Cyber Warfare, IoT Security, Web

Application Security, Operating System Security, Advanced Numerical and Simulation Techniques, Analysis of Stochastic Processes. Eligibility: BS(CE) 4 years Degree Program or equivalent. Requires Graduate Assessment Test (GAT General) scores for admission consideration. Contact: Graduate Coordinator: Dr. Syed Mudassar Hussain (Ext. 411)

Alumnus Testimonial: "It made me what I am today—a place that enabled me to struggle, progress, and work hard for a better life. I'm proud of it!" — Muhammad Jahanzaib 2. Computer Science

Computer science is the study of the theory, experimentation, and engineering that form the

basis for the design and use of computers. It is the scientific and practical approach to computation and its applications and the systematic study of the feasibility, structure, expression, and mechanization of methodical procedures (or algorithms) that underlie the acquisition, representation, processing, storage, communication of, and access to information. Computer Science at BUIEMS

The Department of Computer Science at BUIEMS is one of the pioneer departments that was founded in 2002 as an undergraduate program along with the inauguration of the University. Later, the department expanded its offerings to MS in 2006 and PhD in 2014 respectively. The Computer Science Department of BUIEMS aspires to provide a comprehensive computational education environment that inspires curiosity, creativity and innovation for the enhancement of society in a positive and meaningful way. We

are committed to impart fundamental and highly

technical knowledge on computational issues involving computer communication, networking, security, software development, database management systems, data sciences, and programming. Fact File & Achievements

Code: BSCS

Duration: Four years (8 Semesters) Accreditation: Officially Accredited by the National Computing Education Accreditation Council (NCEAC) and categorized in the highest 'W' category. Rank: Clinched fourth position among top ranking universities of the country in Computer Science.

Program Mission: Our mission is to become intellectual forefronts of computer science that nurtures its students and faculty to address real life problems by applying sound computing knowledge and participating in cutting edge innovative research in scientific, economic and social paradigms.

Scheme of Study (BS Computer Science)

Semester 1: Introduction to ICT, Programming Fundamentals, English Composition & Comprehension, Calculus & Analytical Geometry, Islamic Studies, Applied Physics. Semester 2: Object Oriented Programming, Communication and Presentation Skills, CS

Supporting 1, Discrete Structures, Digital Logic & Design. 3: Data Structures & Algorithms, Software Engineering, Linear Algebra,

Semester

Computer Organization & Assembly Language, University Elective-1. Semester 4: Computer Networks, Operating Systems, University Elective 2, Theory of Automata, CS Supporting 2. Semester 5: Artificial Intelligence, Database Systems, Technical and Business Writing,

Compiler Construction, Probability and Statistics. Semester 6: Track Elective 1, Track Elective 2, Design & Analysis of Algorithms, CS Supporting 3, University Elective 3, University Elective 4 - Social Service. Semester 7: Track Elective 3, Track Elective 4, Parallel & Distributed Computing, Final Year Project-1, Information Security, Professional Practices.

Semester 8: Track Elective 5, Track Elective 6, Final Year Project - II, University Elective 5, Pakistan Studies. BS Program Summary

Total Credit Hours: 133 Essentials/Eligibility: Adheres to the admission criteria/policy of HEC/NCEAC for BS CS

intake. The minimum requirements for admission are at least 50% marks in Intermediate (HSSC) examination with Mathematics, OR an equivalent qualification with Mathematics certified by IBCC. Admission is contingent upon passing the admission test. Contact: Chairperson: Dr. Mehmood Baryalai (Ext. 710) Graduate Program Information (MS Computer Science)

The Department offers the Master of Science (MS) degree in Computer Science. The

Department presently has active research groups in the following areas: Communication Systems Group (CSG), Computational Intelligence Group (CIG), Information Retrieval & Processing (IRP), Wireless & Sensor Networks (WSN), System Security Group (SSG), Distributed & Networked Systems (DNS), and Machine Learning and Big Data. MS Degree Requirements: At least 24 hours of graduate level course work and at least

6 hours Master's Thesis. Total Credit Hours: 30. Core Graduate Courses: Advanced Theory of Computation, Advanced Algorithm Analysis, Advanced Operating System, Advanced Computer Architecture. Elective Domains: Software Engineering, Artificial Intelligence, Information Management, Human Computer Interaction, Computer Science Education, Network Performance

Evaluation. Eligibility: BS(CS) 4 years Degree Program or equivalent. GAT General scores are required. Contact: Graduate Coordinator: Dr. Syed Mudassar Hussain (Ext. 411) Alumnus Testimonial: "I have to admit I was very nervous on my first day but there was no

reason at all to be so. All the students, teachers, and other people I've met, everyone is so friendly, helpful and supportive... there is no pressure either from University works or from the teachers." — Environmental Feedback from Muhammad Fayzan Malik.

3. Electrical Engineering

Electrical engineering is one of the dynamic fields which is continually introducing new technologies to benefit society. Electrical Engineering at BUIEMS

The Department of Electrical Engineering at BUIEMS was founded in 2012, initially offering an undergraduate degree in Electrical Engineering with a focus on Electric Power Systems Engineering. The motivation of the program is to offer quality education opportunities and produce qualified engineers for the growing power sector of the country. The department's undergraduate program has been accredited by the Pakistan Engineering Council (PEC) under

Level-II, which is a recognition of high-quality education standards. To accommodate practicing engineers looking to conduct research, the department initiated a graduate (MS) program in Spring 2014. Programs Offered & Fact File

Programs: BS ELE, MS ELE, PhD ELE.

Code: BSELE Duration: Four years (8 Semesters) Accreditation: Pakistan Engineering Council (PEC) Recognized under Level-II. Program Mission: The department of electrical engineering imparts quality learning in the field of electrical engineering, promotes excellence in research & innovation, and plays a leading role in economic, social, and cultural development.

Program Educational Objectives (PEOs): PEO-01: Demonstrate professional competence in the field of electrical engineering through in-depth knowledge and skills in the area as well as through contemporary interdisciplinary studies. PEO-02: Engage in continual professional development through effective communication,

teamwork, and lifelong learning. PEO-03: Depict professional integrity, practice, and implement ethics, and social responsibility to the global community. Scheme of Study (BS Electrical Engineering)

Semester 1: Reading and Writing Skills, Calculus and Analytical Geometry, Computing

Fundamentals, Islamic Studies/Ethics, Applied Physics, Engineering Drawing. Semester 2: Differential Equations, Linear Circuit Analysis, Workshop Practices, Pakistan Studies, Programming Fundamentals, Electronic Devices and Circuits. Semester 3: Linear Algebra, Communication & Presentation Skills, Digital Logic Design, Complex Variables and & Transforms, Electrical Network Analysis.

Semester 4: Electromagnetic Field Theory, Embedded Systems, Signal & Systems, Probability Methods in Engineering, Social Sciences Elective - I.

Semester

Semester 5: Electrical Machines, Technical Report Writing, Breadth Core - I, Computing Elective, Natural Sciences Elective. Semester 6: Communication Systems, Linear Control Systems, Breadth Core-II, Social Sciences Elective - II, IDEE-I.

7: Depth Elective - I, Depth Elective-II, Depth Elective - III, Management Sciences Elective - I, Senior Design Project-I. Semester 8: Depth Elective-IV, Depth Elective - V, Management Sciences Elective - II, IDEE-II, Senior Design Project - II. Electives Track Distributions

List of Computing Electives: Data Structures and Algorithms, Databases, Artificial Intelligence, Machine Learning, Mobile Application Development, Web Application Development, Software Engineering, Network Security. Electrical Engineering Electives (Power Systems Engineering Stream): Instrumentation and Measurements (Breadth Core I), Power Distribution and Utilization

(Breadth Core II), Power System Analysis, Advanced Electrical Machines, Power Generation, Electrical Power Transmission, Power Electronics, Power System Protection, Power System Operation & Control, Electrical Machine Design and Maintenance, High Voltage Engineering, Renewable Energy Systems, Digital Signal Processing, Industrial Drives, FACTS and HVDC Transmission, Data Communication, Smart Grid. Electrical Engineering Electives (Communication / Telecommunication Engineering

Stream): Computer Communication Networks (Breadth Core I), Electronic Circuit Design (Breadth Core II), Digital Communications, Microwave Engineering, Antennas and Wave Propagation, Digital Signal Processing, Instrumentation and Measurements, Transmission and Switching systems, Wireless and Mobile Communications, Data Communication, Satellite Communication, Optical Communication, RF and Microwave Engineering,

Navigation and Radar Systems, Digital Image Processing, Emerging Wireless Technologies and RF Planning, Telecommunication Policies and Standards, Multimedia Systems, Telecom Traffic Engineering, Broadband and Next Generation Networks, Network Security, Radar Systems Engineering, Telecommunication Network Management.

Interdisciplinary Elective Engineering (IDEE) Courses: Basic Mechanical Engineering, Surveying and Leveling, Operating Systems, VLSI Design, ASIC/FPGA Design, Database Management Systems, Computer Communication Network, Artificial Intelligence, Big Data Analytics, Machine Learning, Computer Vision, Computer Modelling and Simulations,

Thermodynamics, Mechanics of Materials, General Surveying, Construction Materials, Manufacturing Processes, Engineering Statics & Dynamics, Introduction to Environmental Engineering, Environmental Economics, Energy and Environment, Environmental Laws and Policies.

BS Program Summary

Total Credit Hours: 136 Essentials/Eligibility: F.Sc (Pre-Engineering) / ICS with Mathematics & Physics from any recognized board or equivalent with at least 60% marks, OR a Diploma of Associate

Engineer (DAE) in the relevant field securing at least 60% marks. Pass in the university entry test is mandatory. Contact: Chairperson: Engr. Dr. Surat Khan (Ext. 416) MS Program Summary

Total Credit Hours: 30 Essentials: BS(EE) 4 years Degree Program or equivalent. Requires Graduate

Assessment Test (GAT General) scores for admission. Contact: Graduate Coordinator: Dr. Syed Mudassar Hussain (Ext. 411) 4. Electronic Engineering

The Department of Electronic Engineering at BUITEMS is dedicated to addressing future

technological requirements of both industry and society as a whole. Electronic Engineering at BUITEMS

Established in 2003, the department has been at the forefront of providing quality education and training to the youth of Pakistan, empowering them with knowledge of future technologies. Graduates have excelled across various professional domains, finding employment in National

and International Research Institutions, Telecom Companies, Banks, and Multinationals, alongside Automotive, Textile, Biomedical, Petrochemical, and Manufacturing fields. Our students enjoy access to state-of-the-art facilities and laboratories: well-equipped Computer Labs, Basic Electronics Lab, Digital System Lab, Microprocessor Lab, Basic Telecommunication Lab, PCB Lab, Control Automotive and Robotics Lab, Laboratory of Instrumentation & Industrial

Automation, and the Embedded System Lab. Over 400 students get placed under internship programs and industrial study tours every year. Fact File & Programs Offered

Programs: Four-year BS (EE), Two-year MS (EE). Code: BSEE Duration: Four years (8 Semesters) Accreditation: Pakistan Engineering Council (PEC) Recognized.

Program Mission: Our mission is to produce highly skilled engineers who can contribute effectively in the field of electronic engineering by equipping them with analytic skills and innovation required for research, development, and production. Program Educational Objectives (PEOs): PEO-01: Contribute their knowledge for the betterment of humanity in general and society

in particular.

Semester

PEO-02: Impart their analytical and innovation skills for the enhancement of research, development, and production of Electronic Systems. PEO-03: Work efficiently in teams or individually, integrating effectively in diverse cultures and environments.

Scheme of Study (BS Electronic Engineering)

Semester 1: Functional English, Calculus and Analytical Geometry, Linear Circuit Analysis, Applied Physics, Computer Fundamentals & Programming, Workshop Practice. 2: Communication Skills, Islamic

Studies, Engineering Drawing, Pakistan Studies, Linear Algebra, Electronic Devices & Circuits.

Semester 3: Differential Equations, Instrumentation and Measurement, Digital Logic Design, Probability and Random Variables, Electronic Circuit Design. Semester 4: Complex Variables and Transform, Technical Report Writing and Presentation skills, Analog Integrated Electronics, Electrical Network Analysis, Microprocessors and Microcontrollers.

Semester 5: Electrical Machines, Object Oriented Programming for Engineers, Control Systems, Signals and Systems, Data Communication & Networks. Semester 6: Power Electronics, Numerical Analysis, IDEE-I, Communication System, Management Elective-I. Semester 7: EE Elective-I, Social Science Elective-I, EE Elective-II, Management

Elective-II, Industrial Electronics, Final Year Project Part-I. Semester 8: EE Elective-III, EE Elective-IV, Digital Signal Processing, Social Science Elective-II, IDEE-II, Final Year Project Part-II. BS Program Summary

Total Credit Hours: 136 Essentials/Eligibility: F.Sc (Pre-Engineering) / ICS with Mathematics & Physics from

any recognized board or equivalent with at least 60% marks, OR a Diploma of Associate Engineer (DAE) in the relevant field securing at least 60% marks. Must clear the admission test. Contact: Chairperson: Dr. Muhammad Ayub Tareen (Ext. 401) Graduate Program Information (MS Electronic Engineering)

The Department offers the Master of Science (MS) degree in Electronic Engineering. Core Courses: Linear Control, Stochastic Processes. Elective Specialized Domains: Electronics and Embedded Systems: Advanced DSP, Advanced Integrated Electronics, Advanced Computer Architecture, Advanced Industrial Electronics,

Advanced VLSI Design, Nano Electronics, MEMS Sensor Design, 3D Printing, Advanced Opto Electronics, Advanced FPGA Design, Advanced Instrumentation Systems, Multirate Systems And Filter Banks, Advanced Machine Drives, Switched Mode Converter Analysis And Design, Advanced Power Electronics, Advanced Image Processing.

Control Systems: Modelling And Simulations, Multivariable Control, Optimal Control, Nonlinear Control, Digital Control Systems, Advanced Control Systems, Adaptive Control, Stochastic Control, Model Predictive Control. Robotics: Advanced IPC, Intelligent Control, Advanced Topics In Control Systems.

RF Electronics: Advanced Microwave Engineering, Advanced Antenna Design, Advanced RF Electronics, Advanced Electromagnetic Theory, Transmission Lines, Advanced Concepts In RADAR Applications, Microwave-Based Testing, Microwave Integrated Circuit Design. Communication Systems: Advanced Digital Communication, Cellular And Mobile

Communication, Stochastic Processes, Information And Coding Theory, Advanced Topics

Semester

In Communication Systems, Advanced Mobile Networks, Advanced Optical Comm., Advanced Satellite Comm., Radio Wave Propagation. Management Electives: Engineering Management, Research Methodology, Project Management.

MS Summary: Total Credit Hours: 30. Prerequisites include a BS(EE) 4 years degree program or equivalent and a valid GAT General score. Contact: Graduate Coordinator: Dr. Syed Mudassar Hussain (Ext. 411) 5. Information Technology

Information Technology ensures that an organization's systems, networks, data, and applications link up and operate appropriately. An IT professional handles significant areas: deploying and maintaining business applications, services, and infrastructure (servers, networks, storage). Information Technology at BUITEMS

The Department of Information Technology at BUITEMS was founded in 2007 as an undergraduate program and later expanded its offerings to an MS in Information Technology. It features a growing, research-oriented faculty. Ranked by the Higher Education Commission (HEC) as one of the top undergraduate programs in the field of IT in the country, the program prepares students with a sound knowledge base of programming, systems analysis, and

design, business telecommunications, and database management. The term 'information technology' was famously coined by the Harvard Business Review to make a distinction between purpose-built machines and general-purpose programmable computing machines. Fact File & Objectives

Code: BSIT Duration: Four years (8 Semesters)

Accreditation: NCEAC Recognized. Rank: Clinched 4th position nationally in Computer Science and IT criteria. Program Objectives: To give students a sound knowledge base of programming, systems analysis and design, business telecommunications, and database management with concentration in a variety

of areas. To produce graduates who have good grounding and a wide range of knowledge and competence in the sector of information science and technology. To train students for work in production, development, management, and maintenance of technological systems.

Program Mission: The mission of the Information Technology department is to equip students with state-of-the-art tools and technology for the management and analysis of IT infrastructure. Furthermore, it aims to train students to build and maintain effective, reliable, secure, and modern information systems to support organizational functions. Scheme of Study (BS Information Technology)

Semester 1: Introduction to ICT, Programming Fundamentals, English Composition & Comprehension, Basic Mathematics, Applied Physics, Islamic Studies. Semester 2: Object Oriented Programming, Communication & Presentation Skills, Calculus & Analytical Geometry, Pakistan Studies, University Elective-I, Discrete

Structures. Semester 3: Data Structure & Algorithms, Database Systems, Computer Networks, University Elective II, Linear Algebra. Semester 4: Operating Systems, Web Technologies, Information Security, Probability & Statistics, University Elective III.

Semester 5: Technical & Business Writing, System & Network Administration, Software Engineering, IT Infrastructure, DB Administration and Management. Semester 6: IT Supporting I, IT Elective I, Virtual Systems & Services, IT Project Management, IT Elective II. Semester 7: Final Year Project-I, IT Elective III, Cyber Security, IT Supporting II, IT

Supporting III. Semester 8: Final Year Project-II, University Elective IV, IT Elective IV, IT Elective V, Professional Practices. BS Program Summary

Total Credit Hours: 134 Essentials/Eligibility: F.Sc / ICS with Mathematics from any recognized board or equivalent with at least 50% marks, OR a Diploma of Associate Engineer (DAE) in the relevant field securing at least 60% marks. Pass in the admission test is mandatory. Contact: Chairperson: Dr. Bushra Naeem (Ext. 535) 6. Software Engineering

Software Engineering is the application of a systematic, disciplined, and quantifiable approach to the design, development, operation, and maintenance of software systems. Software Engineering at BUITEMS

The Department of Software Engineering at BUITEMS was founded in 2012 as an undergraduate program with just three faculty members and fifty students. Today, the Software

Engineering department has eight faculty members, most of whom are foreign qualified from the best universities in the world. The pedagogy employed is 'learning by doing' through practical exercises, activity-driven lecturing, real-life projects, and soft skills training. Fact File & Objectives

Code: BSSE

Duration: Four years (8 Semesters) Accreditation: Pakistan Engineering Council (PEC) Recognized. Program Vision: To attain excellence in education and research at the national and international level and produce technically, professionally, and ethically sound graduates.

Program Mission: To produce next-generation Software Engineering graduates who are fully equipped with concrete software engineering knowledge, real-world problem-solving skills, research abilities, excellent leadership qualities, and entrepreneurial abilities required for their future careers, and serving their communities.

Program Educational Objectives (PEOs): PEO-01: Utilize critical-thinking and engineering abilities to excel in the domain of Software Engineering. PEO-02: Act as professional, ethical, and knowledgeable citizens. PEO-03: Demonstrate leadership and work collaboratively in diverse teams/organizations.

Scheme of Study (BS Software Engineering)

Semester 1: Information and Communication Technology (ICT), Software Engineering Fundamentals, Reading and Writing Skills, Applied Physics, Psychology, Islamic Studies and Ethics, Occupational Health and Safety. Semester 2: Programming Fundamentals, Communication & Presentation Skills, Discrete

Structures, Calculus & Analytical Geom., Web Development (WD), Pakistan Studies and Global Perspective. Semester 3: Object-Oriented Programming, Probability & Statistics, Computer Architecture and Logic Design, Linear Algebra, SE Elective I (Web Engineering (WE)). Semester 4: Data Structure & Algorithms, Software Design & Architecture, Human- Computer Interaction, SE Elective - II, Principles of Management, Complex Variables and

Transform. Semester 5: Software Construction and Development, Entrepreneurship and Marketing, SE Elective - III, Database Systems, Operating Systems. Semester 6: Software Quality Engineering, Computer Networks, Numerical Analysis, MDEE-I, Technical & Business Writing.

Semester 7: Software Project Management, Information Security, Senior Design Project- I, SE Elective - IV, Design and Analysis of Algorithms. Semester 8: MDEE-II, Senior Design Project-II, SE Elective-V, SE Elective-VI*, Formal Methods in Software Engineering. Elective Course Lists

Software Engineering Electives: Software Engineering Economics, Mobile Application Development, Visual Programming, Information System Audit, Real-Time Systems, Graphic Designing, Big Data Analytics, Cloud Computing, Semantic Web, Systems Programming, Agent-Based Software Engineering, Management Information Systems, Data Encryption and Security, Natural Language Processing, Software Metrics, E-

Commerce, Machine Learning, Artificial Intelligence, Data Visualization, Computer Graphics, Software Testing, Computer Vision, Simulation and Modeling.

Multidisciplinary Engineering Electives (MDEE): Embedded Systems, Internet of Things (IoT), Blockchain, Image Processing, Game Application Development, Multimedia Communications, Data Encryption and Security. BS Program Summary

Total Credit Hours: 132 Essentials/Eligibility: F.Sc (Pre-Engineering) / ICS with Mathematics & Physics from any recognized board or equivalent with at least 60% marks, OR a Diploma of Associate Engineer (DAE) in the relevant field securing at least 60% marks. Admission is contingent upon qualifying the admission test.

Contact: Chairperson: Dr. Shah Marjan (Ext. 412) Faculty Profiles (FICT)

FICT has a remarkable interdisciplinary faculty working in a wide range of disciplines ranging across Computing, Embedded Systems, Communications and Wireless Networks, Electrical and Power Engineering, and Technology Management. Notable features include a high degree

of scholarly global collaboration. Professors

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Associate Professors

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Support: We Care for Our Students

Support to the students in every possible way is central to BUITEMS in general and FICT in particular. We take your career seriously, so we work to incorporate qualities and abilities that employers require into your experience at the university, whether you're studying, volunteering,

playing sports, or taking part in work placement programs. Hierarchy of Support Services

Class Counselor: Class counselor is your first stop when you need any assistance. At

FICT every class has a class counselor whose duties include supporting and helping students of his or her class in all kinds of academic and non-academic issues. They advise you on your subjects, sometimes arrange supervisors for you, and help you in planning your progress. **Departmental Focal Person:** Every department has one focal person (also called CMS

focal person) who is responsible for course registrations/enrollments, student records, results, and all other issues related to the automated Campus Management Solution (CMS). **Chairperson's Office:** The Chairperson's office is available for assistance if you need help beyond the class counselor and the focal person's responsibility. The Chairperson

and staff are available during office hours to listen to your problems. The office can also be accessed via email and telephone. The counselor and focal person may themselves consult the Chairperson's office to resolve your problem or query. The Chairperson and his or her office are always there to provide you full support with a smile. **Faculty Coordinator:** The Faculty Coordinator overlooks academic and non-academic matters at the overall level of the Faculty of Information and Communication Technology.

Occasionally, you might need to consult the Faculty Coordinator in case the Chairperson's office refers your request / problem further. One-to-one consultations can be scheduled to plan successful learning avenues. **Dean's Office:** The Dean's office is the highest office at the Faculty of Information and Communication Technology. Characterized by instant support with a smile, the office

sows and spreads courtesy and student support throughout FICT. The office assists students in resolving problems that require the help of the Registrar's office, the Finance section, and the office of the Controller of Examinations, when the problem cannot be resolved at the Chairperson's or the Faculty Coordinator's office. **International Students Coordinator:** Programs offered at FICT have been popular

among international students hailing from diverse nationalities. FICT has an International Students Coordinator dedicated to helping international students at the faculty. The international students coordinator helps students with a wide range of general assistance and support on issues that relate to international student admissions, cultural adjustment, academics, community relations, and campus-wide assistance.

Specialized Framework Support

Disabled Students: BUITEMS extends extra support to students with disabilities. The infrastructure at BUITEMS is developed keeping in consideration the needs of disabled students. The building facilities of

FICT have the necessary access features for disabled students. Students with specific learning difficulties, physical/mobility, or visual

impairments are fully supported; when needed, instructions and evaluations are adapted to the needs of disabled students. Financial Support: We boast our financial support at BUIITEMS where every third student receives financial assistance in the form of scholarships and fee waivers. The

students receive scholarships and fee waivers under a wide range of programs, backed by the university as well as with the support of our friends including USAID, PPL, and many more. BUIITEMS also offers a work and study program that not only supports students with a full fee waiver but also provides exposure to work environments. The office of financial assistance at BUIITEMS lets you know more about the available options.

Career Services: We want you to be a success, both now and in the future. From the moment you arrive at BUIITEMS, we can help you enhance your CV, build your experience, and network with potential employers. The mission of the BUIITEMS Career Unit is to provide advice, services, programs, and resources to empower students and alumni throughout their career development journey. We partner with potential employers

to offer internship and job opportunities. Services Offered: Career Assessment, Career Advising/Counseling, Workshops and Panels, Job Listing/Job Placement, Mock Interviews with Feedback, On-Campus Interviewing/Recruitment, Internship Program, and Career Fairs. Contact Support Team: 081-2880511 or kinza.khan@buitms.edu.pk

Travelling to BUIITEMS & Distances

Quetta is the capital of the province of Balochistan. Being one of the important cities of Pakistan, Quetta is easily accessible by all modern means of transportation. By Car / Bus: If you plan to travel to Quetta by road, you can take major national highways leading to Quetta. The city is connected to Karachi, which is 686 km away, via national highway N25. Quetta is connected to Lahore, 935 km away via N70 and N5, and

980 km via N50. Quetta is 911 km from Islamabad via N50 and 835 km from Peshawar via N50 and N55. Major bus service companies operate routes to Quetta from all major cities in the country. By Air: BUIITEMS is located at a distance of 3 km from Quetta International Airport. The airport, served by various national and international carriers, offers continuous

connections to other major cities in Pakistan and international destinations. By Train: Quetta Railway Station is one of the major railway stations in the country. The railway track, established in the 1890s during the British era, connects Quetta with the rest of the country. The extensive network of Pakistan Railways connects Quetta to Karachi in the south by an 863 km track, Lahore in the northeast (1,170 km), and

Peshawar further northeast (1,587 km). Regular train services connect Quetta to the rest of the country. Verified Inter-City Distances to BUIITEMS

Karachi to BUIITEMS: 686 km

Lahore to BUIITEMS: 935 km Islamabad to BUIITEMS: 911 km Peshawar to BUIITEMS: 835 km Multan to BUIITEMS: 586 km

Hyderabad to BUIITEMS: 712 km